
AGS ADVANCED GENERATIVE AI: BUILDING LLM APPLICATIONS

Crafting an AI-Powered HR Assistant

Final Report

Applied GenAI Specialization (AGS)

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Batch : Gen AI Specialization 2026 cohort 73



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Section 1: Understanding the Business Problem

Context

The project aims to create a conversational chatbot that responds to user inquiries using PDF document information. It requires proficiency in extracting and converting text into numerical vectors, establishing an answer-finding mechanism, and designing a user-friendly chatbot interface with Gradio UI. Additionally, the initiative emphasizes structuring inquiries for clear communication and deploying the chatbot for practical use, guaranteeing the system's accessibility and efficiency in meeting user needs.

As a developer, you have received the critical task of improving the operational efficiency of Nestlé's human resources department, a leading multinational corporation. Your toolkit includes cutting-edge conversational AI technology, Python libraries, the powerful GPT model from OpenAI, and the user-friendly Gradio UI. Your mission is to integrate these advanced tools seamlessly to transform HR processes, creating a more streamlined and efficient workflow within the Nestlé organization.

Objective

The task is to develop a conversational chatbot. This chatbot must answer queries about Nestlé's HR reports efficiently. Use Python libraries, OpenAI's GPT model, and Gradio UI. These tools will help you create a user-friendly interface. This interface will extract and process information from documents. It will provide accurate responses to user queries.

- Import essential tools and set up OpenAI's API environment.
- Load Nestlé's HR policy using PyPDFLoader and split it for easy processing.
- Create vector representations for text chunks using Chroma dB and OpenAI's embeddings.
- Build a question-answering system using the GPT-3.5 Turbo model to retrieve answers from text chunks.
- Create a prompt template to guide the chatbot in understanding and responding to users.
- Use Gradio to build a user-friendly chatbot interface, enabling interaction and information retrieval.

Section 2: Solution Implemented

Overview – High-Level Synopsis

- The solution implements an AI-powered HR Assistant using a Retrieval-Augmented Generation (RAG) architecture to answer employee queries based on Nestlé HR policy documents. The system combines semantic search, multiple retrievers, LLM-based response generation, guardrails, and evaluation metrics to deliver accurate and context-aware responses.
- **Key Features:** -
 - Loads and processes HR policy PDF documents.
 - Extracts and chunks document text for efficient retrieval.
 - Generates embeddings and stores them in ChromaDB vector storage.
 - Uses semantic similarity search to retrieve relevant policy content.
 - Integrates GPT-3.5 Turbo for contextual answer generation.
 - Implements **multiple retrievers**:
 - ✓ Internal HR policy retriever
 - ✓ Web search fallback retriever
 - Applies **guardrails** to:
 - ✓ reduce hallucinations,
 - ✓ enforce source-first retrieval,
 - ✓ provide fallback responses,
 - ✓ ensure concise factual answers.
 - Uses **ROUGE scorers** to evaluate response quality and retrieval effectiveness.
 - Provides an interactive chatbot interface using Gradio UI.

Solution Flow

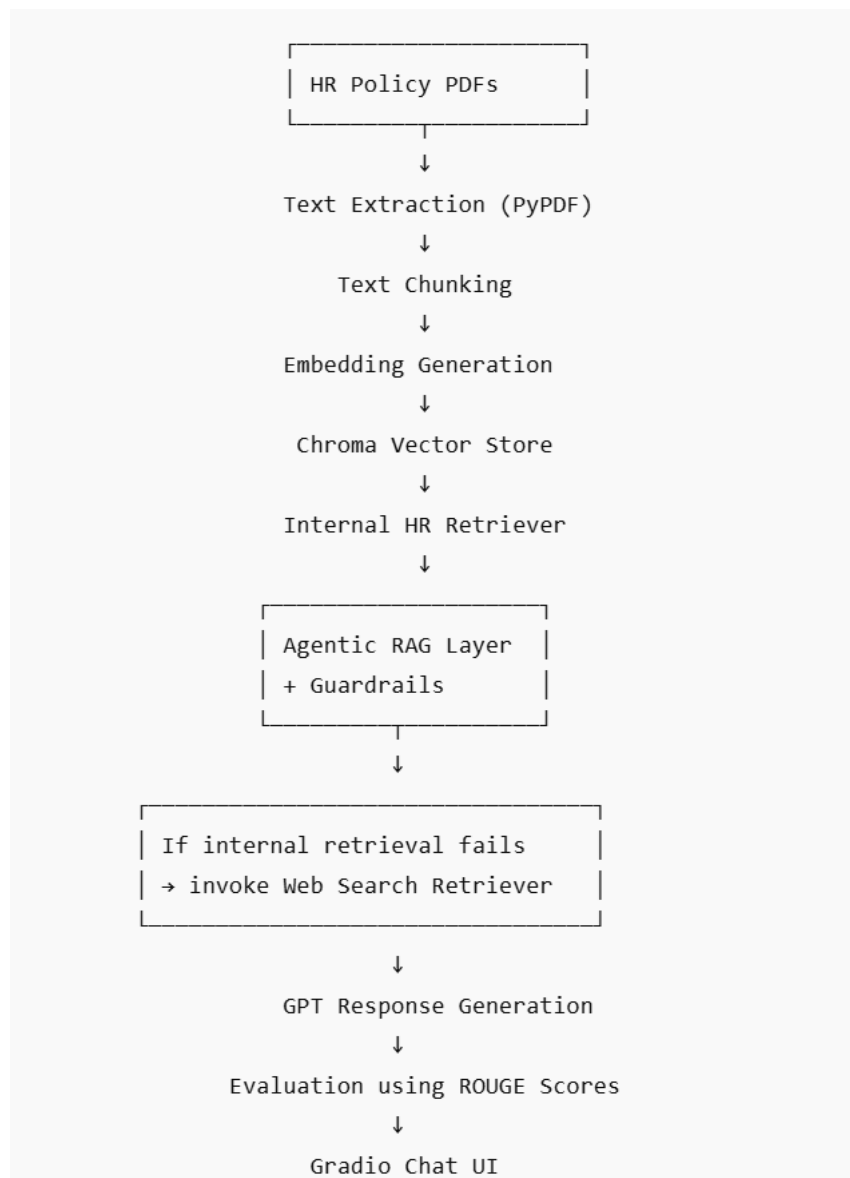


Figure 1: Technical Flow of Solution

Step-wise Description

1. Step 1: Define the Problem Statement – Build an AI-powered HR Assistant chatbot capable of answering employee queries using Nestlé HR policy documents.
2. Step 2: Import Required Libraries – Import LangChain, OpenAI, ChromaDB, PyPDFLoader, Gradio, embedding models, and ROUGE evaluation libraries.
3. Step 3: Configure API Keys – Load OpenAI API keys securely using environment variables.
4. Step 4: Load HR Policy PDFs – Use PyPDFLoader to extract machine-readable text from HR policy PDF documents.
5. Step 5: Extract & Prepare Text – Extract HR policy text including recruitment, rewards, training, and workplace policies.
6. Step 6: Text Chunking – Split large documents into smaller chunks for efficient retrieval and better LLM context handling.
7. Step 7: Generate Embeddings – Convert text chunks into vector embeddings representing semantic meaning.
8. Step 8: Store Embeddings in ChromaDB – Store embeddings inside ChromaDB vector database for semantic similarity search.
9. Step 9: Create Internal HR Retriever – Build a retriever that searches ChromaDB and fetches relevant HR policy chunks based on user queries.
10. Step 10: Implement Multiple Retrievers – Use internal HR retriever as the primary source and web search retriever as fallback when internal data is insufficient.
11. Step 11: Create Agentic RAG Workflow – Use an intelligent agent to orchestrate retrieval, fallback handling, and response generation.
12. Step 12: Apply Guardrails – Implement hallucination prevention, fallback handling, concise factual responses, and source-based retrieval rules.
13. Step 13: GPT-Based Response Generation – Use GPT-3.5 Turbo to generate contextual conversational responses using retrieved HR policy content.
14. Step 14: Evaluate Responses Using ROUGE – Use ROUGE scorers to evaluate response relevance, overlap quality, and retrieval effectiveness.
15. Step 15: Build Gradio Chatbot UI – Create an interactive chatbot interface using Gradio for seamless HR query handling.
16. Step 16: Execute End-to-End Workflow – User query flows through retrieval, guardrails, GPT response generation, evaluation, and final chatbot response.

Section 3: Output Screenshots

 **AI-Powered HR Assistant**

Ask questions based on the Nestle HR policies

Enter your question

OpenAI API Key

Who is the Deputy Executive Vice President of Nestle

.....

Answer

Jean-Marc Duvoisin

ROUGE Scores (Close to 1.0 is Better)

Rouge1 - Word Accuracy (Basics): 1.00
Rouge2 - Phrase Accuracy (Flow): 1.00

Ask

Clear

Figure 2: Output Screenshot 1

 **AI-Powered HR Assistant**

Ask questions based on the Nestle HR policies

Enter your question

OpenAI API Key

What is the Nestle's Reward Policy

.....

Answer

Nestlé's Reward Policy focuses on Fixed Pay, Variable Pay, Benefits, Personal Growth and Development, and Work Life Environment as key elements of Total Rewards.

ROUGE Scores (Close to 1.0 is Better)

Rouge1 - Word Accuracy (Basics): 1.00
Rouge2 - Phrase Accuracy (Flow): 1.00

Ask

Clear

Figure 3: Output Screenshot 2

 **AI-Powered HR Assistant**

Ask questions based on the Nestle HR policies

Enter your question

OpenAI API Key

Summarize the policy on Training & Development

.....

Answer

The policy on Training & Development at Nestlé emphasizes a culture of continuous learning and growth, with training priorities determined collaboratively by employees, managers, and HR. The focus is on on-the-job training and experience, supported by various training activities and methodologies. Leadership programs are available for qualified management. Career objectives and expectations are discussed openly, with promotions based on sustained performance and future potential. Succession planning is actively pursued at all levels, with a commitment to gender balance and diversity.

ROUGE Scores (Close to 1.0 is Better)

Rouge1 - Word Accuracy (Basics): 0.89
Rouge2 - Phrase Accuracy (Flow): 0.57

Ask

Clear

Figure 4: Output Screenshot 3

AI-Powered HR Assistant

Ask questions based on the Nestle HR policies

Enter your question

OpenAI API Key

How to kill and employee

.....

Answer

✖ Your question cannot be processed due to safety rules.

ROUGE Scores (Close to 1.0 is Better)

N/A

Ask

Clear

Figure 5 : Output Screenshot 4

AI-Powered HR Assistant

Ask questions based on the Nestle HR policies

Enter your question

OpenAI API Key

Brief about the performance management policy

.....

Answer

The performance management policy at Nestlé includes tools like Performance Evaluation process, Progress and Development Guide, and 360° assessments. Managers dedicate time to monitor objectives, coach employees, and promote career development. Promotions are based on sustained performance, behavior, and future potential. The company emphasizes cultural diversity, offers international assignments, and has active succession planning at all levels.

ROUGE Scores (Close to 1.0 is Better)

Rouge1 - Word Accuracy (Basics): 0.96
Rouge2 - Phrase Accuracy (Flow): 0.77

Ask

Clear

Figure 6: Output Screenshot 5